ESMS-Net: Enhancing Semantic-Mask Segmentation Network With Pyramid Atrousformer for Remote Sensing Image

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***Abstract*— Transformers has gained widespread adoption in remote sensing image (RSI) segmentation. However, RSI has densely overlapping terrain and significant shadow, making it challenging to segment the blended boundaries of terrains that are the hard classes. Currently, most transformer-based methods construct the self-attention with a sliding window, which influences the feature receptive fields to conquer the intersecting and overlapping objects. Additionally, they often rarely focus specifically on the representation of these hard segmentation objects. To overcome these challenges, we propose a novel Enhancing Semantic Mask Segmentation Network (ESMS-Net) framework including a local–global joint encoder, an auxiliary enhanced encoder, and a multiscale dense decoder. In the local–global joint encoder, we construct a Pyramid Pooling AtrousFormer (PPAFormer) that performs the self-attention with a pyramid-structured atrous sliding window, which enhances the range of receptive fields and the global representation performance. Meanwhile, we construct the dual-feature fusion module (DFFM) and multilevel feature weighted fusion (MFWF) in the multiscale dense decoder to reduce information loss and facilitate the interaction of deep semantic information. For the auxiliary enhanced encoder, we develop a semantic mask based on the predicted results to maintain the hard segmentation classes, and then use the same structure as the first two stages of the local–global joint encoder to learn the hard regions again. Extensive experiments demonstrate the proposed ESMS-Net can achieve significant improvements for segmentation performance compared with the state-of-the-art methods on the ISPRS-Vaihingen and Potsdam datasets. The code will be available at https://github.com/Wzysaber/ESMS-Net.**

***Index Terms*— Pyramid atrous structure, remote sensing image (RSI), semantic mask, semantic segmentation, transformer.**

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I. Introduction

With the rapid development of sensor and aerospace technology, more and more remote sensing (RS) images (RSIs) are collected to acquire the information of earth surface [1], [2]. A basic problem for RSIs is to analyze the categories of earth surface. Semantic segmentation is a significant technique to discriminate the categories, which has been applied in a wide range of fields, including vehicle recognition [3], forest conservation [4], road detection [5], and urban planning [6]. Semantic segmentation aims to implement pixel-level classification of images, enabling to obtain the land-cover type of each pixel. RSIs exhibit diversity, complexity, and class uneven distribution of categories, making semantic segmentation more challenging.

For RSI semantic segmentation, convolutional neural networks (CNNs) have yielded remarkable results. The fully convolutional network (FCN) [7] utilizes encoder–decoder structure with end-to-end training to achieve pixel-level segmentation, reaching the highest accuracy at that time. Due to the excellent segmentation performance of FCN, the encoder–decoder structure has become a primary architecture to construct semantic segmentation models. However, during the decoding process of FCN, the decoder loses a significant amount of feature information, thus necessitating the integration of semantic information from different levels as comprehensively as possible. Subsequently, UNet [8] incorporates skip connection to fuse features from both deep and shallow layers for reducing information loss. In a similar vein, Deeplab V3+ [9] utilizes encoder–decoder structure and skip connection to achieve improvement. It leverages a pyramid structure [10] with dilated convolutions to extract deep semantic features at different scales and integrates the shallow-level features from decoder. To learn rich multiscale semantic features, a lot of multiscale models have been proposed such as multilayer feature fusion [11], and spatial pyramid pooling [12]. Convolutional kernel focuses on pixels within a receptive field to extract information, primarily

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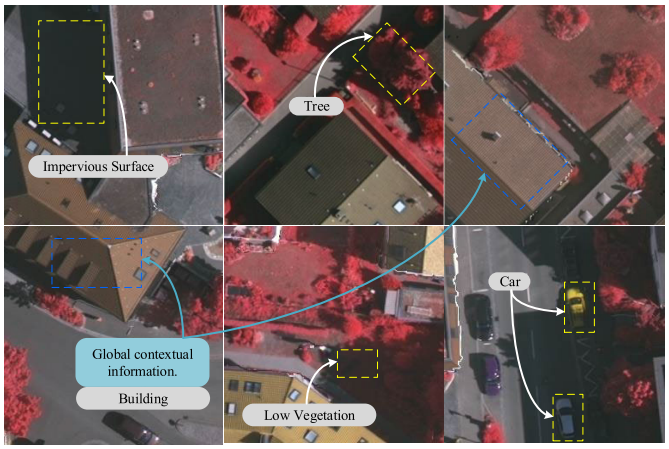


Fig. 1. RSIs from the Vaihingen dataset. The land-covers such as "Car," "Low Vegetation," and "Tree" are usually obstructed by other land-covers or shadows, while the same land-cover category, such as "Building," is distributed in noncontiguous regions.

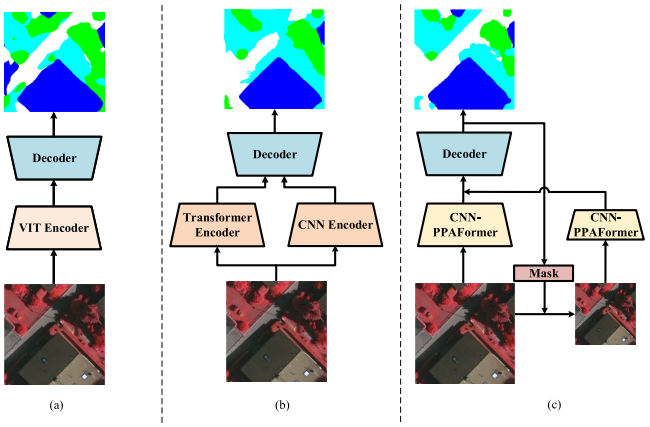


Fig. 2. Different transformer frameworks for semantic segmentation of RSIs. (a) Traditional Transformers [21], [22], [23] directly utilized as backbones to extract characteristic information from RSIs. (b) Parallel encoder structure comprising both Transformers and CNNs [24], [25] has shown to achieve optimal results. (c) We propose a unique universal framework, involving a CNN-transformer structure and a dynamic mask to process hard segmentation classes according to segmentation results.

capturing local details. Thus, the CNN models tend to learn local features during the feature extraction process.

Since RSIs often present characteristics such as high similarity [13] between different categories, mutual occlusion of objects [14], and significant shadow-covered areas. As depicted in Fig. 1, in RSIs, multiple land cover categories obstruct each other, which poses a significant challenge for accurate segmentation. Therefore, the global contextual information [15] is very important to understand RSIs and improve segmentation accuracy. To aggregate global contextual information, Transformers [16] gradually demonstrate their superiority. The Vision Transformer (ViT) [17] divides images into fixed-size patches and then treats them as sequences to aggregate global information. The Swin Transformer [18] employs a shifting window without overlapping to reduce the computational cost, allowing these window patches to interact and construct global information. The CrossFormer [19] introduces cross-scale embedding layer and long-short distance attention to achieve cross-scale self-attention. The BiFormer [20] employs dynamic sparse attention to perform flexible computation allocation and content awareness, endowing it with dynamic query-aware sparsity. These Transformers capture global feature information through self-attention to effectively establish long-range dependencies in images, thereby breaking through local limitation of convolutional operation. However, these existing Transformer frameworks have limitations when applied to RSI due to the dense and complex terrain features, overlapping, and shadow coverage. Most Transformer methods use window-based approaches that perform self-attention within local regions, which often limits the receptive field of features and overlooks the importance of scale in RSIs. Furthermore, these methods do not consider the feature learning of hard terrains.

To solve the above problems, we propose a novel RSI semantic segmentation network, called Enhancing Semantic-Mask Segmentation Network (ESMS-Net). As shown in Fig. 2, our framework, compared with previous Transformer architectures, builds a suitable CNN-PPAFormer structure for complex RSIs and focuses specifically on hard classes via masking. We design a Pyramid Pooling AtrousFormer (PPAFormer) structure after the convolution layer to form the local–global joint encoder branch, which can simultaneously capture the local information and global properties at different scales. Additionally, a dual-feature fusion module (DFFM) is developed to facilitate the fusion of cross-scale deep semantic information, effectively mitigating occlusion and enhancing the discriminating performance of land-cover features in segmentation. After that, a semantic mask is constructed based on the initial predicted segmentation map. This mask can filter out the large regions with a large number of land-cover class to focus on the small regions with a small number of land-cover classes in the original images. Based on the reserved regions that are challenging to segment, we design an auxiliary enhanced encoder branch to enhance the land-cover feature representation from the small regions, which has the same structure as the first two stages of local–global joint encoder branch. In the decoder, we have developed a multiscale dense decoder structure that combines the local–global features and the enhanced features. To prevent information loss and fuse more effective semantic features, we propose a multilevel feature weighted fusion (MFWF) block to fuse the multiscale dense decoder features. MFWF combines the shallow and deep information at different levels to achieve rich feature aggregation for RS segmentation.

The main contributions of this article are as follows.

1) To represent the global properties of RSIs, we design a PPAFormer block, which consists of the self-attention mechanism in the form of an atrous pyramid structure. This better achieves the interaction of long-range information from different regions.
2) To further enhance the interaction of deep semantic features, we propose a DFFM to fuse the features from the last two stages in the local–global joint encoder branch based on PPAFormer and residual block. It constructs

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a multiscale feature attention (MSFA) block to match land-cover distribution with different shapes.

3) To highlight the hard-to-segmentation regions with small size, we develop an auxiliary enhanced encoder branch based on the predicted segmentation results to construct a semantic mask. This can further enhance the feature representation of small land-cover areas to improve the segmentation performance of RSIs.
4) To further enhance segmentation refinement and effectively fuse features from different stages, we employ a densely connected decoding approach. Then, we develop a MFWF block to fuse the multilevel features of the last decoding layer, which enriches the diversity of features to enhance the discriminative power of different land-covers.

The rest organization of this article is in the following. Section II gives the related works. The details of the proposed method is shown in Section III. Section IV analyzes the experimental results and discusses the effectiveness of the network. In Section V, some conclusion is summarized.

II. Related Works

A. CNN-Based RSI Segmentation

CNNs have been extensively utilized in semantic segmentation of RSIs. FCN [7] was initially introduced in semantic segmentation, employing an end-to-end manner for segmentation and gaining prominence. However, it does not fully integrate the deep and shallow features. Subsequently, UNet [8] was introduced to extract deep semantic information through different downsampling layers in the encoder and then restore spatial resolution with a symmetric convolutional decoder structure. Additionally, skip connections were employed to reduce information loss. In the subsequent development of semantic segmentation models, UNet structure has played significant roles. However, simple skip connections are not fine-grained enough for segmentation and lack the ability to establish relationships between individual pixels. To effectively achieve more fine-grained features, Deeplabv3+ [9] introduced the atrous spatial pyramid pooling (ASPP) module. ASPP uses dilated convolutions with different dilation rates to capture contextual information from different spatial scales. This extracted information is integrated into the shallow levels in the decoder of Deeplabv3+, enhancing spatial features and significantly improving segmentation performance. HRNet [26] achieves the parallel multiscale fusion of high-to-low resolution convolutions to facilitate strong and spatially precise high-resolution representations, effectively mitigating the information loss during downsampling.

Due to the various complex land-cover features and the blurred edge information in RSIs, it is essential to integrate local features and perceive global information in network models. ABCNet [27] designs a bilateral pathway structure to capture global contextual information and a feature aggregation module to fuse contextual features. In addition, local edge details are also crucial information in RSIs, and blurry edge features will restrict the accuracy of segmentation. ERN [28] has an edge loss enhancement module via spatial boundary context to reduce semantic ambiguity. MCCANet [29] enables dynamic and adaptive feature fusion in a context-aware manner, reducing intraclass variations and enhancing boundary contours.

B. Attention-Based RSI Segmentation

In extracting image features, CNN has limitations in perceiving the entire information of image. To interact information from different regions, attention mechanisms are introduced to simulate global contextual information. SENet [30] adopts global average pooling to facilitate channel interaction and autonomously learn the weights associated with different channels. CBAM [31] simultaneously employs spatial attention and channel attention to adaptively refine features, enhancing the feature extraction ability of the network model without significantly increasing computational complexity and network parameters. DANet [32] introduces position attention and channel attention, which represents local features while enriching context-related information.

To address the problem of limited receptive field of CNNs, researchers have incorporated attention mechanisms at various locations in the network structure to efficiently simulate global feature information. MANet [15] employs multiple efficient attention modules to extract contextual dependencies and establish long-range feature map dependencies. SPANet [33] constructs successive pooling attention, effectively improving small-scale object segmentation and object boundaries in RSIs. Meanwhile, to alleviate the computational demands of attention and obtain a larger range of global feature information, researchers construct attention mechanisms on shallow information and multiple scales. A2-FPN [34] enhances multiscale feature learning via attention aggregation module and constructs a pyramid structure to aggregate deep semantic information across multiple scales. TANet [35] incorporates an attention feature enhancement module to globally enhance shallow-level features. However, the attention models in the above methods mainly rely on dot-product and linear mechanisms, overly depending on convolution operations for extracting features from RSIs, thus limiting the representation of global features. Faced with intricate land cover scenes in RSIs, the attention mechanism still lacks the adequate capability to extract global feature information.

C. Transformer-Based RSI Segmentation

In recent years, Transformer models have obtained increasingly widespread application in the field of computer vision, achieving significant advancements, particularly in semantic segmentation tasks. Convolutional operations tend to extract local features, while Transformers excel in mining long-range dependencies and effectively capturing global context information. ViTs have demonstrated their effectiveness in segmentation tasks by adopting a sequence-to-sequence modeling approach. Currently, most Transformers for semantic segmentation are based on the encoder–decoder structure. ViT [17] and Swin Transformer [18] have achieved excellent results in vision tasks, synchronously promoting the increasing application of Transformers in semantic segmentation.

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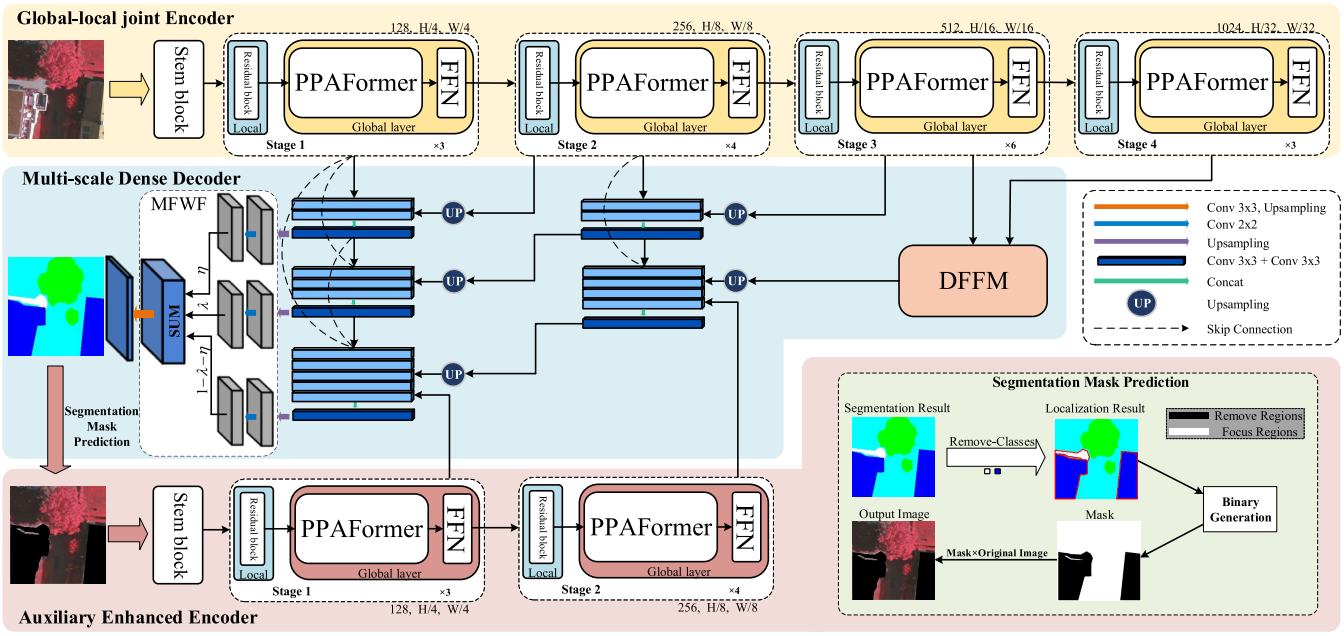


Fig. 3. Architecture of ESMS-Net. ESMS-Net includes three branches: local–global joint encoder to learn local–global features, auxiliary enhanced encoder to highlight the feature representation of hard segmented land-covers, and multiscale dense decoder to fuse the features from different scales. The dynamic Mask updates based on the predicted segmentation results in each iteration, emphasizing the focal points that the model finds difficult to learn in each iteration.

DC-Swin [22] employs Swin Transformer as a feature extractor to capture multiscale global contextual features, then it designs a densely connected feature aggregation module to fuse the extracted multiscale information. SegViT [36] introduces an attention-to-mask module, which leverages an attention to generate semantic segmentation masks.

While Transformers excel at effectively capturing global features, CNNs have a distinct advantage in extracting local information. Therefore, it is essential to integrate CNNs and Transformers for utilizing their respective strengths. STransFuse [37] adopts a dual-encoder architecture to separately extract features from both CNN and transformer encoders, which constructs an adaptive fusion module to effectively extract local and global information. UNetFormer [38] introduces convolutional operations into self-attention to aggregate both local and global information. CMT [39] employs CNN to extract local information while integrating lightweight multihead self-attention (MHSA) to further capture global features. TopFormer [40] takes tokens from various scales as inputs to generate scale-aware semantic features, which are then injected into the corresponding tokens for improving representation.

III. Method

A. Overall Architecture

The overall architecture of ESMS-Net, as shown in Fig. 3, utilizes the Residual Block from ResNet50 [41] and the proposed PPAFormer, constructed with a pyramid structure, to design the local–global joint encoder structure. The Residual Block extracts local information through convolutional layers, while PPAFormer extracts global contextual information through self-attention. The fusion of local and global information enables more precise feature representation.

For a given RSI $F \in \mathbb{R}^{(3\times H\times W)}$, we should comprehensively represent its features to reveal the intrinsic of each class land-cover in the encoder. Thus, the Residual Block and PPAFormer are employed in a sequential manner to fuse local and global information, which can reflect the global and local properties of land-covers to generate feature maps $f_i \in \mathrm{R}^{C_i\times(H/2^{i+1})\times(W/2^{i+1})}$, where $i \in \{1, 2, 3, 4\}$. To fully mine the abstract and essential features of RSIs, there is four stages with different depths in the encoder. The network depths for different stages are represented as depth $\in \{3, 4, 6, 3\}$ and $C_i \in \{128, 256, 512, 1024\}$. In addition, in the decoder, we first use the stem block with convolution and pooling operation to reduce the scale of input image for obtaining low computation cost.

In RSI, different land-covers have uneven distributed areas. The land-covers in small regions are easy to be interfered by other land-covers and difficult to be segmented. Therefore, we add an auxiliary enhanced encoder branch, possessing the same structure of the first two stages in the local–global joint encoder branch, to enhance the features of hard segmented land-covers. In this branch, we make full use of the predicted semantic information to generate a mask, which is binarized to filter the easily segmented land-covers. The filtered new images are directed into the auxiliary enhancement branch to extract feature information from hard-to-segment regions. The auxiliary enhanced encoder enriches and supplements the features that might have been lost in the local–global joint encoder, ultimately facilitating finer differentiation of hard land-covers.

In the decoder, the multiscale dense connection [42] is used to aggregate different features, which can fully combine the features from different scales and reduce information loss in the decoder. The features from the local–global joint encoder

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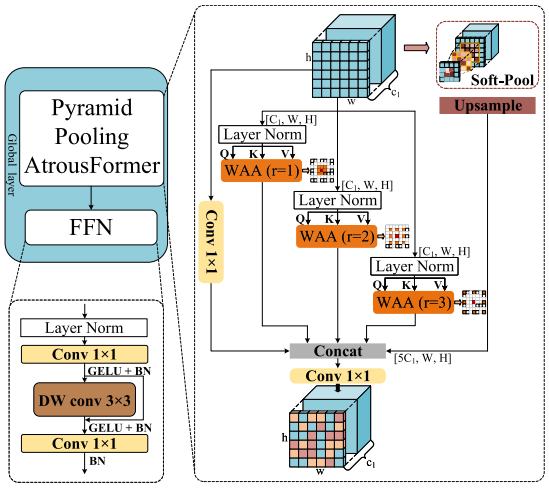


Fig. 4. Structure of the PPAFormer. A pyramid structure is designed by employing three WAA with different dilation rates. Each WAA utilizes a sliding window of size $d \times d$, centered at position (i, j), to acquire self-attended keys and values, enabling the interaction of information across different scales of receptive fields. Additionally, it maximizes the fusion of original features through Conv 1×1 and Soft-Pool.

and the auxiliary enhanced encoder are densely connected to achieve maximum information fusion. Then, the dense features are used to generate the predicted segmentation map. However, only using single-level dense features will waste the output features from the other levels. Therefore, we propose a MFWF method, which fully combines these features from different levels with different weight coefficients to achieve more refined segmentation outcome. Additionally, in the third and fourth stages of the encoder, we propose DFFM to perform cross-fusion decoder of deep semantic information from different scales, thereby enhancing adaptability to land-covers with varying shapes.

B. PPAFormer

To fully capture the global information, we propose a PPAFormer block based on the proposed Window Atrous Attention (WAA) to perform self-attention with different receptive fields. WAA applies a sliding window of size $d \times d$ centered at position (i, j) to obtain the key and value for self-attention. To obtain the information from different receptive fields, we use multiple WAA branches with different atrous rates in the same window size to form the pyramid structure.

For each WAA branch, the atrous rate, denoted as $r \in \mathbb{N}^+$, is used to control the receptive field size. To enhance the diversity of features in each WAA branch, the channels of the feature maps are averagely divided into several groups to form the MHSA structure, where each head adopts the same atrous rate. Following this, the features from different heads are combined to enhance the diversity of features and then input into a linear layer. The output y_{ij}^r of WAA at position (i, j) is defined as

$$\begin{aligned} y_{ij}^r &= \mathrm{WAA}\left(q_{ij}, K_{ij}^r, V_{ij}^r\right) \\ &= \prod_h \left[\mathrm{Softmax}\left(\frac{q_{ij}\left(K_{ij}^r\right)^T}{\sqrt{d_k}}\right) V_{ij}^r\right]_h \end{aligned} \tag{1}$$

where $1 \le i \le W$, $1 \le j \le H$. d_k is the dimension of each head. $\prod$ is the concatenation operate to concatenate the multihead attentions. K_{ij}^r and V_{ij}^r represent the keys and values within a $d \times d$ window at position (i, j). For the query at position (i, j), the corresponding window position (i', j') will be used as the keys and values for the self-attention operation, and the window position is

$$\left\{ \left(i', j'\right) | i' = i + m \times r, j' = j + n \times r, -\frac{d}{2} \le m, n \le \frac{d}{2} \right\}. \tag{2}$$

Therefore, the key, query, and value of input X can be obtained with a 1×1 convolution operation $f^{1\times1}$ as follows:

$$q_{ij} = f^{1\times1}(X)_{i'j'},\ K_{ij}^r = f^{1\times1}(X)_{i'j'},\ V_{ij}^r = f^{1\times1}(X)_{i'j'}. \tag{3}$$

WAA conducts the self-attention across all the position with a sliding window to achieve the attention of the whole image. We use different atrous rates in WAA to construct a pyramid structure [43], as illustrated in Fig. 4. The PPAFormer achieves larger receptive fields for self-attention on the feature maps by concatenating the parallel WAA with different atrous rates. In addition, we also use the Soft-Pool and the residual connection with a 1×1 convolution to preserve the information as much as possible at various scales. By concatenating these parallel branches to form the PPAFormer module, this approach constructs the long-range dependencies with a sliding window to obtain the global information for enhancing the semantic features.

For an input feature map $S' \in \mathbb{R}^{(C_1 \times H \times W)}$, we use the atrous rates $r = 1, 2, 3$, and the outputs of WAA and 1×1 convolution are

$$F_1 = \mathrm{Concat}\left(\mathrm{WAA}_{r=i}\left(S'\right),\ i = 1, 2, 3\right) \tag{4}$$

$$F_2 = \mathrm{Conv}_{1\times1}\left(S'\right). \tag{5}$$

For the Soft-Pool [44], it dynamically weights the feature map based on the importance of features. This approach enables to adaptively capture the significant information by highlighting the key areas. The Soft-Pool effectively preserves feature information to a maximum extent and restores it to the original size through bilinear interpolation during upsampling. The Soft-Pool and the output of corresponding branch are defined as follows:

$$\text{Soft-Pool}(u_i) = \sum_{i \in R} \frac{e^{u_i} \cdot u_i}{\sum_{j \in \mathrm{R}} e^{u_j}} \tag{6}$$

$$F_3 = \mathrm{Upsampling}\left(\text{Soft-Pool}\left(S'\right)\right). \tag{7}$$

For a feature map, R is a pooling window. After applying the Soft-Pool to S', the pooling features can be obtained, and then it is further upsampled with bilinear interpolation to restore its initial size. Finally, the features F_1, F_2, and the upsampled features F_3 are concatenated to obtain the pyramid pooling features. After that, we use a 1×1 convolution to recover the original size. Finally, we use the FFN block to further enhance the interaction and representation of semantic features. The output feature map L is defined as

$$L = \mathrm{FFN}(\mathrm{Conv}_{1\times1}(\mathrm{Concat}(F1,\ F2,\ F3))). \tag{8}$$

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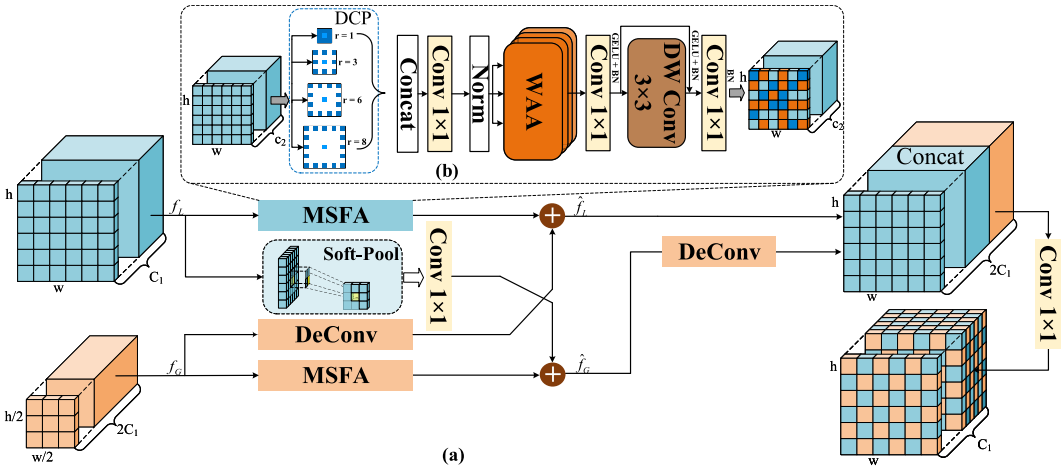


Fig. 5. Structure of DFFM. (a) We cross-fuse deep semantic information using DFFM. (b) MSFA enhances adaptability to various land cover shapes by further expanding the receptive fields.

C. Dual-Feature Fusion Module

To better integrate the deep-level semantic information, we propose a novel DFFM to fuse the features from the third and fourth stages of the local–global joint branch. As illustrated in Fig. 5, we construct a MSFA to aggregate the feature information from different scales, enhancing the adaptability of the land-covers with different shapes and further expanding the receptive field of features. Based on the MSFA, we implement the interaction of deep semantic features to improve the segmentation performance.

Similar to the ASPP [43], we construct a dilated convolutional pyramid (DCP) to capture the features from various receptive fields. The optimal dilation rate is set to $(r = 1, 3, 6, 8)$ for better representing the multiscale features. Then, the proposed WAA with $r = 1$ and FFN are used to highlight the importance features based on the long-dependencies. We can obtain the MSFA block by combining the DCP, WAA, and FFN. By fusing the spatial local information from DCP with different dilated scales and the global feature information from WAA with a dilation rate $(r = 1)$, we further expand the receptive fields, capturing deep-semantic information to match different land-cover shapes.

To achieve the interaction of deep semantic information from the third and fourth stages of the local–global joint encoder, the features are denoted as f_L and f_G. The channel dimension of f_G is twice that of f_L, while the spatial size of f_G is half that of f_L. We use the deconvolution (DeCov), the Soft-Pool, and 1×1 convolution to unify the channel and the spatial scale for the feature interaction as follows:

$$\begin{aligned} \hat{f}_L &= \mathrm{MSFA}(f_L) \oplus \mathrm{DeConv}(f_G) \\ \hat{f}_G &= \mathrm{SoftPool}(f_L) \oplus \mathrm{MSFA}(f_G). \end{aligned} \tag{9}$$

The features extracted from stages 3 and 4 are cross-fused using MSFA to obtain $\hat{f}_L$ and $\hat{f}_G$. $\hat{f}_G$ is then upsampled to match the feature map size of $\hat{f}_L$ using deconvolution, through a 1×1 convolution for fusion after concatenation. The interaction features are fused to the large scale in the following:

$$L_D = \mathrm{Concat}\left(\hat{f}_L, \mathrm{DeConv}\left(\hat{f}_G\right)\right) \tag{10}$$

$$L = \mathrm{Conv}_{1\times1}(L_D). \tag{11}$$

D. Auxiliary Enhanced Encoder

For the hard segmentation land-covers, we construct an auxiliary enhanced encoder branch to learn the hard classes alone. This branch has the same structure as the first two stages in the local–global joint encoder. We utilize a mask to remove the easy segmentation land-covers. The mask is generated from the binarized segmentation results and dynamically adjusted according to each training.

In RSIs, some land-cover classes with large sample size enable to effectively extract features for models. Conversely, small sample sizes and irregular land-cover classes present significant challenges, including mutual occlusion, shadow coverage, and blurred edges, which make accurate segmentation difficult for the model. Therefore, these areas are defined as hard-to-segment regions. We categorize different regions of the image into easily segmented areas and hard-to-segment regions based on the predicted categories. Specifically, if the predicted results belong to hard categories, the corresponding regions are considered hard-to-segment regions, which we retain for relearning.

As shown in Fig. 3, the generated mask is applied to the original input image. In the Vaihingen and Potsdam datasets, the large sample size classes, i.e., "Impervious Surface" and "Building," are considered easy-to-segment land-covers. To focus on the hard segmentation land-covers, we leverage the prediction segmentation results from the global–local joint encoder to identify the two classes of land-covers and generate corresponding masks for their removal. The remained hard segmentation land-covers are again used for feature learning through the auxiliary enhanced encoder branch, which can enhance the discriminating performance of hard classes.

The predicted map is denoted as $P \in \mathbb{R}^{(C_1 \times H \times W)}$, where each channel corresponds to the prediction probability for a specific class. To obtain the predicted label of each pixel, we use the argmax operation along the channel axis. The predicted label map $P_c \in \mathbb{R}^{(H \times W)}$ has the same scale as the original input image $(H \times W)$. Based on the predicted

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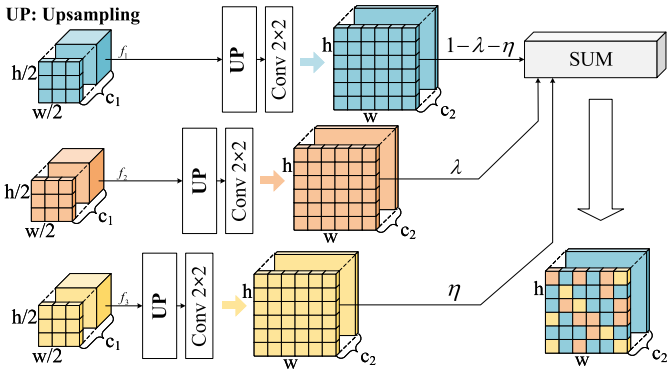


Fig. 6. Structure of MFWF. In the multiscale dense decoder, multiple branches are weighted and fused to improve semantic information interaction and reduce information loss.

label map, we construct the binary mask $B_k \in \mathbb{R}^{(H\times W)}$ in the following formula:

$$B_k(i,j) = \begin{cases} 1, & l(i,j) \in P_h \\ 0, & \text{Otherwise} \end{cases} \tag{12}$$

where $l(i,j)$ represents the predicted label at the spatial coordinate (i,j). When the prediction value at position (i,j) is in the hard classes P_h, the corresponding position in the binary mask B_k is set to 1, otherwise 0. We use two smaller number of land-covers as the hard classes in this article. Then, the binary mask B_k is multiplied with the original input image $L \in \mathbb{R}^{(C_2\times H\times W)}$ in an element-wise manner. The masked image T_k is represented as

$$T_k = L \odot B_k. \tag{13}$$

After obtaining the masked images, they are input into the auxiliary enhanced encoder, which can enhance the representation performance of hard classes and refine the semantic features. Finally, the masked-semantic features and the local–global features are fused in the decoder, improving the segmentation accuracy of RSI.

E. Multilevel Feature Weighted Fusion

To fully fuse features at different scales and enhance feature representation, we employ dense connections in the decoders to facilitate feature interaction, allowing different semantic information to achieve cross-scale interaction. The obtained dense single-level features can improve the segmentation performance of RS. However, the designed model contains different levels, with the single-level features, resulting in the waste of the other level features in the proposed network. Thus, we design a MFWF method to integrate the semantical information from various levels with weighting fusion. As depicted in Fig. 6, after feature fusion via the Multiscale Dense Decoder, we can obtain three level features f_1, f_2, and f_3. Then, all the three levels are upsampled to the original image size $(H \times W)$ with ConvTranspose2d, and the feature maps $S' \in \mathbb{R}^{(C_1\times H\times W)}$ are processed with a 2×2 convolution to change the number of channels to C_2. The formula is as follows:

$$S_i' = \text{Conv}_{2\times 2}(\text{Upsampling}(f_i)), \quad n = 1, 2, 3. \tag{14}$$

Among the three feature levels, the feature level f_1 contains most of the fused feature information. After restoring the scale, S_1' is the main feature level, which integrates the deep encoding features. The remaining feature levels, S_2' and S_3', are used to aggregate the intermediate and shallow encoding features, respectively. To constrain the contribution of each level, we set two parameters λ and η to achieve the balance of each level features. The weighted summation operation is used to combine the three level features as follows:

$$L_D = (1-\lambda-\eta)\cdot S_1' + \lambda \cdot S_2' + \eta \cdot S_3'. \tag{15}$$

After obtaining the fused multilevel feature maps L_D, they are processed through convolution and upsampling operation to generate the ultimate segmentation map. With the multilevel feature fusion, the network model reduces the feature loss and combines the diverse features from different network depths to enhance the segmentation precision.

IV. Experiments and Analysis

We evaluated our ESMS-Net model on the International Society of Photogrammetry and Remote Sensing (ISPRS) Vaihingen [45] and Potsdam datasets [45] compared with some recent advanced works. Additionally, we conducted ablation experiments to analyze the effectiveness and contributions of each module, including PPAFormer, DFFM, the auxiliary enhanced encoder, and MFWF.

A. Datasets

The Vaihingen dataset [45] consists of 33 real orthophoto (TOP) images captured by advanced airborne sensors, covering an area of approximately 1.38 km^2 in Vaihingen. The ground sampling distance (GSD) for these images is approximately 9 cm. Each TOP image is composed of three channels: infrared (IR), red (R), and green (G). Following ST-Unet [24] and SCG-TransNet [23], the images with IDs 1, 3, 5, 7, 13, 17, 21, 23, 26, 32, and 37 were used for training, while the images with IDs 11, 15, 28, 30, and 34 were used for testing. All the images were cropped into patches with size of 256 × 256.

The Potsdam [45] consists of 38 images, each with a size of 6000 × 6000 pixels. These images were extracted from a very high-resolution TOP mosaic with a GSD of 5 cm. The dataset covers an area of 3.42 km^2 in the Potsdam and contains complex buildings and dense settlement structures. Similarly, for testing, we selected the images with IDs 2_13, 2_14, 3_13, 3_14, 4_13, 4_14, 4_15, 5_13, 5_14, 5_15, 6_14, 6_15, and 7_13, The remaining images were used for training. The original images were cropped into patches with size of 256 × 256.

B. Implementation Details

The network model is implemented using PyTorch framework. During training, we use the SGD optimizer with a momentum item of 0.9. The initial learning rate is set to 0.01, and we employ the "Poly" decay strategy. All experiments are conducted on an NVIDIA GeForce RTX 3090 24-GB GPU. The batch size is set to 4, and we employ random scaling

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TABLE I
COMPARISON OF SEGMENTATION RESULTS ON THE VAIHINGEN DATASET

Method	IoU (%)					Evaluation index	
	Impervious Surface	Building	Low Vegetation	Tree	Car	MIoU (%)	Average F1 (%)
FCN	73.22	78.97	54.80	70.38	39.92	63.46	76.65
UNet	72.91	81.68	57.23	71.63	48.29	66.35	79.13
Deeplab V3+	74.85	83.01	56.09	71.54	50.3	67.16	79.71
UperNet	73.45	81.50	55.65	71.31	47.26	65.84	78.69
DANet	73.54	81.40	56.88	71.21	42.68	65.14	78.00
TransUNet	73.27	81.01	55.07	71.08	55.13	67.11	79.86
Swin-Unet	69.31	73.37	49.48	67.12	30.78	58.01	72.02
UNetFormer	82.56	81.59	56.79	71.33	54.42	69.34	81.51
CMT	83.16	82.35	57.78	70.67	57.59	70.31	82.12
A^2-FPN	83.31	82.01	58.26	71.79	57.57	70.63	82.32
ST-UNet	76.36	**82.98**	57.79	**72.53**	61.48	70.23	82.15
SCG-TransNet	74.59	82.69	58.54	71.53	**64.10**	70.29	82.27
ESMS-Net	**83.41**	82.55	**60.02**	72.31	60.06	**71.66**	**83.08**

and flipping to process the training images. Additionally, to achieve optimal model performance, we set the maximum number of epochs to 300. To train the proposed network on the Vaihingen and Potsdam datasets, we utilize a joint loss function, including Dice loss and Cross-Entropy (CE) loss, and the joint loss is denoted as

$$\text{Loss} = L_{\text{CE}} + L_{\text{Dice}}. \tag{16}$$

To evaluate the segmentation results, we selected the Overall Accuracy (OA), the mean Intersection over Union (MIoU), and the Average $F1$ score ($F1$) as performance metrics. Based on the confusion matrix, which includes four terms: true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN), the formulas for calculating OA, $F1$ ($F1$ Score), and Intersection over Union (IoU) are as follows:

$$\text{OA} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \tag{17}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{18}$$

$$F1 = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \tag{19}$$

where precision = TP/(TP + FP) and recall = TP/(TP + FN). For segmentation task with multiple classification, MIoU represents the mean IoU for all categories, and Average $F1$ represents the $F1$ score across all categories.

C. Comparison With Other Methods

We compared the segmentation results of ESMS-Net with the latest state-of-the-art methods, including CNN models such as FCN [7], Unet [8], Deeplab V3+ [9], UperNet [46], and DANet [32]. DANet incorporates attention mechanisms into the convolutional network. TransUNet [47], SwinUNet [48], UNetFormer [38], and A^2-FPN [34]. CMT [39] serially connects CNN and Transformer. SCG-TransNet [23] utilizes Transformers as the primary encoder for feature extraction. ST-Unet [24] adopts a parallel mode that combines both transformer and CNN for feature extraction. In contrast, ESMS-Net consists of PPAFormer, DFFM, MFWF, and an auxiliary enhanced encoder, which combines both CNN and transformer to integrate the local and global information and fuses the features from different stages and levels for enhancing the discriminative power of features.

1) Results on the Vaihingen Dataset: Table I illustrates that ESMS-Net exhibits superior performance on the Vaihingen dataset in MIoU and average $F1$ compared with the other methods. Compared with FCN, U-Net integrates spatial information from different stages through skip connections to obtain better results. However, the segmentation performance of U-Net is not as satisfactory as Deeplab V3+. Because Deeplab V3+ adopts a Spatial Pyramid Pooling module to aggregate multiscale global context information, which also shows significant superiority over UperNet and DANet. Among these CNN-models, Deeplab V3+ exhibits the best segmentation performance. In comparison with Deeplab V3+, our ESMS-Net shows a 4.5% improvement in MIoU and a 3.37% improvement in average $F1$.

In the Transformer-models, despite the superior global modeling capability of Transformer, a simple application of Transformer is not effective for RSI segmentation, such as Swin-Unet and UnetFormer. A sequential stacking of convolutional layers and Transformers in TransUnet improves the segmentation accuracies compared with Swin-Unet to demonstrate the effectiveness of combining convolutional and Transformer architectures. ESMS-Net, a backbone composed of stacked CNN-PPAformer, achieves an improvement over TransUnet, showcasing a 4.52% increase in MIoU and a 3.22% increase in average $F1$. CMT concatenates and fuses the Transformer information with local CNN features, whereas A2FPN inserts the Transformer only at the final part of the feature extraction process. ST-Unet adopts a parallel fusion of Transformer and CNN, while SCG-TransUnet introduces multiscale Transformers in the decoder to extract global information. From the experimental results, ESMS-Net outperforms the suboptimal models with an increase of 0.1% and 1.48% in IoU for the "Impervious Surface" and "Low Vegetation" categories, respectively. This demonstrates that ESMS-Net performs better compared with the other Transformer-models in aggregating global context information and establishing long-term dependencies.

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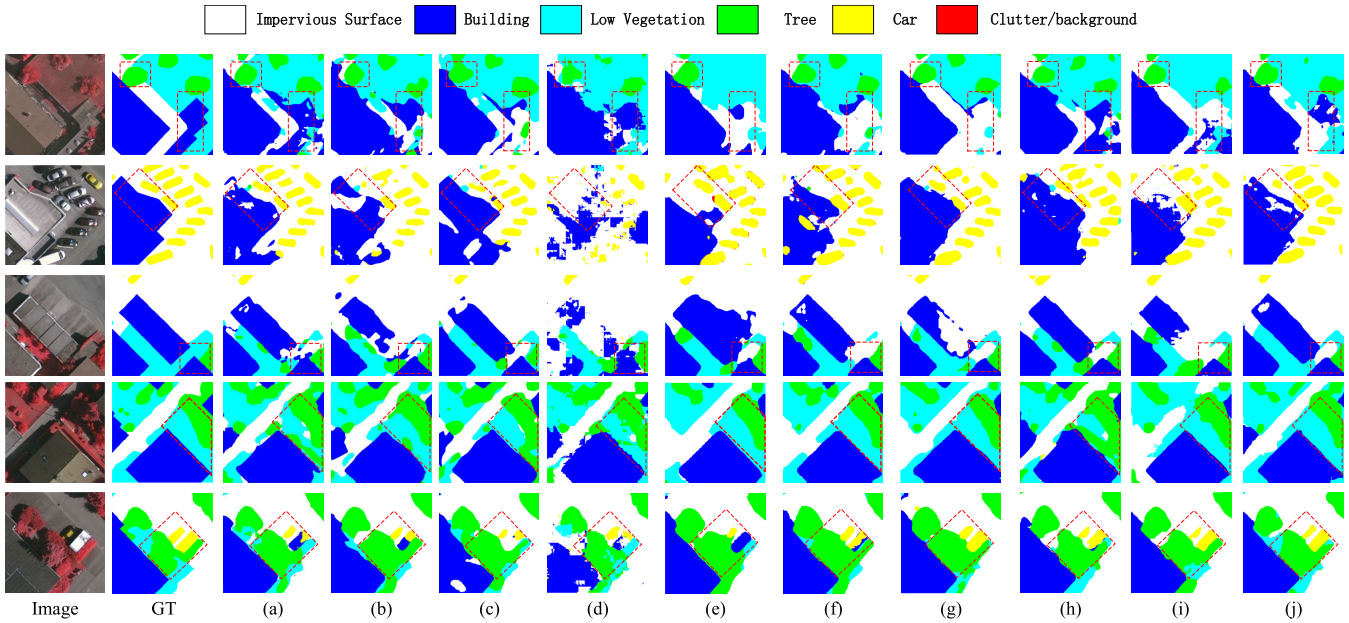


Fig. 7. Visualized results of different networks on the Vaihingen dataset. (a) UNet. (b) Deeplab V3+. (c) TransUNet. (d) Swin-UNet. (e) UnetFormer. (f) CMT. (g) ESMS-Net. (h) A^2-FPN. (i) SCG-TransNet. (j) ESMS-Net.

TABLE II
COMPARISON OF SEGMENTATION RESULTS ON THE POTSDAM DATASET

Method	IoU (%)					Evaluation index	
	Impervious Surface	Building	Low Vegetation	Tree	Car	MIoU (%)	Average F1 (%)
FCN	77.41	83.52	66.10	63.19	74.34	72.91	84.12
UNet	77.10	82.83	64.59	65.44	76.16	73.22	84.35
Deeplab V3+	79.01	84.76	67.53	63.05	78.05	74.48	85.13
UperNet	76.95	83.93	65.65	60.40	76.57	72.70	83.91
DANet	77.35	83.45	66.46	63.47	75.28	73.20	84.32
TransUNet	78.61	85.60	67.16	64.10	79.33	74.96	85.44
Swin-Unet	71.45	75.02	59.03	50.96	71.15	65.52	78.79
UNetFormer	81.59	83.58	67.95	66.61	72.80	74.51	85.27
CMT	81.86	83.91	69.06	**69.38**	75.78	76.00	86.24
A^2-FPN	82.27	85.01	68.27	67.31	73.6	75.31	85.8
ST-UNet	79.19	**86.63**	67.89	66.37	79.77	75.97	86.13
SCG-TransNet	78.41	86.11	68.07	67.65	**79.93**	76.04	86.20
ESMS-Net	**83.92**	86.21	**69.93**	68.87	74.96	**76.78**	**86.72**

To visually illustrate the differences and superiority of our proposed method compared with several semantic segmentation methods, Fig. 7 presents segmentation prediction results for various methods. U-Net and Deeplab V3+ exhibit numerous instances of inaccurate segmentation. Swin-Unet neglects the local spatial information, resulting in fragmented segmentation output. ST-Unet and SCG-TransNet effectively reduce segmentation errors, but there still exist notable segmentation errors, particularly in the edge connection regions between objects. Compared with the other models, ESMS-Net achieves better segmentation details, particularly excelling in the identification of challenging objects such as overlapped and shadow-obscured objects. In the fifth row, ESMS-Net achieves excellent recognition of objects obscured by the shadow of a large vehicle and effectively identifies the connection between the large vehicle and the edge of tree. Additionally, in the fourth row, ESMS-Net shows significant recognition performance in distinguishing the intertwining of "Low Vegetation" and "Tree."

2) Results on the Postdam Dataset: Table II further demonstrates the superiority of our proposed model in terms of segmentation performance. On the Postdam Dataset, U-Net achieves effective segmentation for sparsely distributed categories like "Tree" by fusing feature information at different stages through skip connections in the decoder stage. Compared with Unet, Deeplab V3+ has an improvement of 1.26% in MIoU and 0.78% in $F1$. Deeplab V3+ also achieved the best segmentation results among CNN models. Deeplab V3+ exhibits significantly higher accuracy in identifying "Impervious Surface" and "Building" categories compared with the other convolutional models, which captures more comprehensive global information. Compared with Deeplab V3+, ESMS-Net shows a 4.91% increase in IoU for the "Impervious Surface" and a 1.45% increase in IoU for the "Building."

In Transformer-models, our model perform the best in aggregating global contextual information and establishing long-term dependencies. SCG-TransNet integrates Swin

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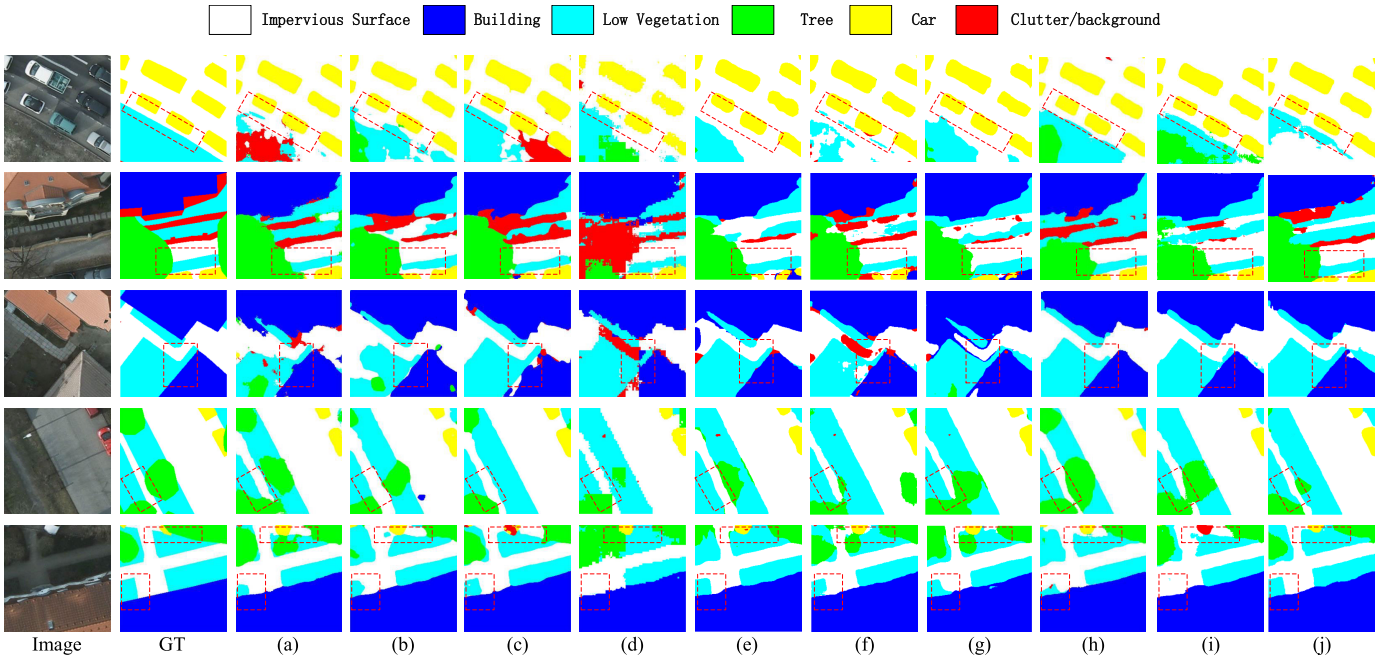


Fig. 8. Visualized results of different networks on the Postdam dataset. (a) UNet. (b) Deeplab V3+. (c) TransUNet. (d) Swin-UNet. (e) UnetFormer. (f) CMT. (g) ESMS-Net. (h) A^2-FPN. (i) SCG-TransNet. (j) ESMS-Net.

transformer with multiscale window sizes in the decoder stage, achieving more accurate segmentation for densely clustered land cover categories such as "Low Vegetation" and "Tree." Compared with SCG-TransNet, ESMS-Net shows a 4.91% increase in IoU for the "Low Vegetation" and a 1.45% increase in IoU for the "Tree." Meanwhile, compared with the suboptimal model, our model has an improvement of 0.74% in MIoU and 0.48% in $F1$. And ESMS-Net achieves a 1.65% improvement in IoU for "Impervious Surface" compared with their suboptimal results.

The visualization of segmentation results for each model is presented in Fig. 8. U-Net's overall framework is end-to-end, and it fuses feature information through skip connections. The majority of CNN models also adopt an end-to-end approach for semantic segmentation. However, in the segmentation results, CNN models display notable errors during the segmentation process. On the other hand, Transformers can achieve more comprehensive and accurate segmentation of land-covers. In the fifth row, having three land covers inter-overlap, the other models often encounter misclassifications and incomplete recognition, while ESMS-Net accurately identifies the overlap and connection between objects. Similarly, in the second row, the shadows from trees simultaneously obscure large areas of "Impervious Surface" and "Low Vegetation," which leads to misidentification and incomplete segmentation of land-covers. ESMS-Net effectively segments the overlapping edges of "Tree" and "Impervious Surface," while accurately identifying "Low Vegetation" even under shadow occlusion. Our proposed ESMS-Net exhibits more precise segmentation performance, especially in recognizing shadow and overlapping objects, which effectively identifies the hard segmentation classes and segments the object boundaries.

D. Ablation Experiments

To validate the effectiveness and rationality of each module within ESMS-Net, we conducted ablation experiments on the Vaihingen dataset. The ESMS-Net model consists of four modules: PPAFormer, DFFM, MFWF, and the auxiliary branch. By removing each module, we can assess the contribution of each component, as shown in Table III and Fig. 9. Models 1 to 4 do not consider the MFWF and only use the final single level outputs as the segmentation results. Model 5 replaces PPAFormer in ESMS-Net with traditional MHSA to further demonstrate its effectiveness.

1) Effectiveness of PPAFormer: We excluded the PPAFormer component from Model 4 in the local–global joint encoder and utilized only the ResBlock from ResNet-50 in the backbone network. The remained model relies on the convolutional layers to learn segmentation features. From Table III, the introduction of PPAFormer, in comparison with Model 4, has improved MIoU by 1.89% and average $F1$ by 1.35%. It has achieved significant improvements with a 1.84% increase in the "Impervious Surface" and a 1.61% increase in the "Building." Additionally, comparing the experimental results of Model 5 with ESMS-Net further demonstrates that PPAFormer outperforms traditional Transformers in segmenting RSIs with unique terrains. PPAFormer not only enhances segmentation accuracy for large-area objects but also achieves more precise segmentation for hard classes like "Low Vegetation" and "Car."

From Fig. 9, there is a significant improvement in the overall segmentation accuracy. In the third column, the overall performance of "Building" is improved, and the misclassification of "Tree" is greatly reduced. The results indicate that PPAFormer can extract global information and establish long-range dependencies.

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TABLE III
ABLATION EXPERIMENTS OF THE PROPOSED MODULES ON THE VAIHINGEN DATASET

Model	Method	IoU (%)					Evaluation index	
		Impervious Surface	Building	Low Vegetation	Tree	Car	MIoU (%)	Average F1 (%)
1	Auxiliary+DFFM	81.82	81.68	57.61	70.51	56.89	69.70	81.69
2	PPAFormer+Auxiliary	83.43	83.05	59.17	**72.42**	58.34	71.29	82.84
3	PPAFormer+DFFM	83.61	82.93	59.26	70.73	59.20	71.15	82.72
4	PPAFormer+Auxiliary+DFFM	**83.68**	**83.29**	59.73	71.94	59.29	71.59	83.04
5	MHSA+Auxiliary+DFFM+MFWF	82.90	82.25	58.32	72.42	58.28	70.84	82.59
ESMS-Net	PPAFormer+Auxiliary+DFFM+MFWF	83.41	82.55	**60.02**	72.31	**60.06**	**71.66**	**83.08**

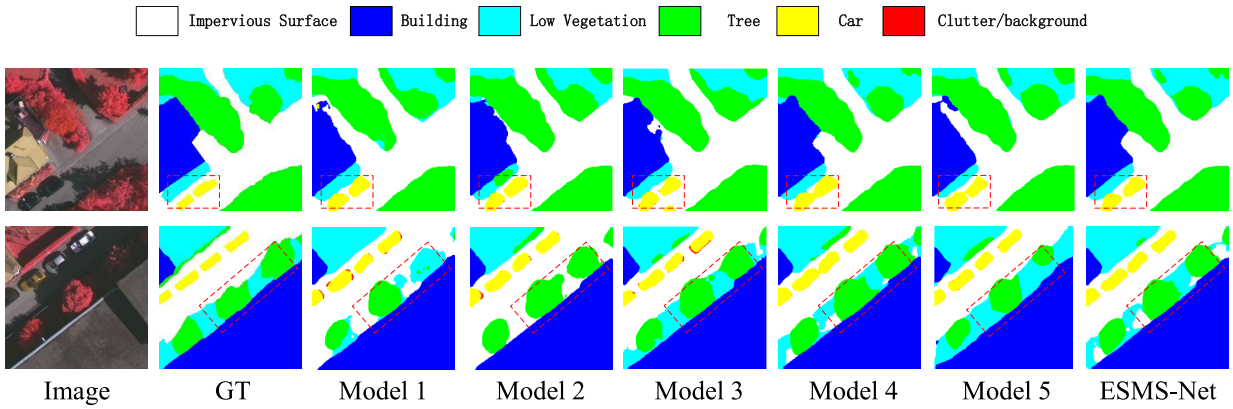


Fig. 9. Visualized results of Ablation experiments on the Vaihingen dataset. (Model 1) Auxiliary+DFFM. (Model 2) PPAFormer+Auxiliary. (Model 3) PPAFormer+DFFM. (Model 4) PPAFormer+Auxiliary+DFFM. (Model 5) MHSA+Auxiliary+DFFM+MFWF.

2) Effectiveness of DFFM: We removed the DFFM module from Model 4 and employed basic upsampling and concatenation methods, followed by convolution, to combine features from stages 3 and 4 of the local–global joint encoder. From Table III, DFFM is incorporated in Model 4, leading to a 0.3% improvement in MIoU and a 0.2% increase in average $F1$. Simultaneously, it can be observed from the segmentation maps in the fourth column that the segmentation accuracy of the yellow "Car" is significantly improved, showing more detailed edge and local information. DFFM achieves cross-fusion of deep semantic information through MSFA and interaction, effectively preserving local detailed information.

3) Effectiveness of Auxiliary Enhanced Encoder: We directly removed the auxiliary enhanced encoder from Model 4 during training. As shown in Table III, the auxiliary enhanced encoder improves the IoU for all categories. It can be observed from the segmentation maps that the overall refined segmentation is enhanced, making it easier to distinguish the similar categories and the hard segmentation classes. MIoU increases over 0.44%, and average $F1$ improves over 0.32%. This demonstrates that the auxiliary enhanced encoder, with segmentation mask prediction, can further learn the features of hard segmentation classes by excluding easily discriminated categories, leading to finer segmentation.

4) Effectiveness of MFWF: Compared with Models 4 and ESMS-Net, introduces MFWF into Model 4, which improves the feature representation via fusing different level features in Table III. The segmentation accuracy of object categories with small samples, such as "Tree" and "Car," improves over 0.37% and 0.77%, respectively. Simultaneously, in the seventh column segmentation maps, the edge of the yellow "Car" is effectively discriminated. MFWF accomplishes more effective fusion of semantic information from different decoding branches, thereby preventing information loss.

TABLE IV
ANALYSIS OF THE WEIGHT PARAMETERS IN MFWF

Model	Weight index		Evaluation index		
	λ	η	OA(%)	MIoU (%)	Average F1 (%)
MFWF(λ,η)	0.1	0.1	85.55	71.41	82.90
	0.1	0.2	85.76	71.57	83.07
	0.2	0.1	**85.78**	**71.66**	**83.08**
	0.2	0.2	85.74	71.46	82.90

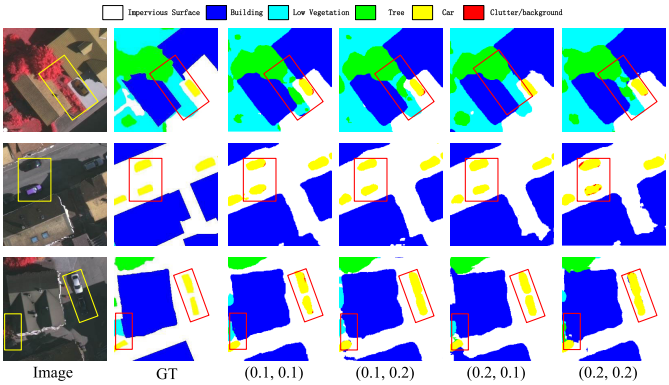


Fig. 10. Visualized results under different weight parameters in MFWF on the Vaihingen dataset.

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TABLE V
RESULTS FOR MASK UNDER DIFFERENT REMOVED CATEGORIES ON THE VAIHINGEN DATASET

Excluded category	Impervious Surface	Building	Low Vegetation	Evaluation index		
				OA(%)	MIoU (%)	Average F1 (%)
num=1	✓			85.37	71.12	82.76
		✓		85.41	71.07	82.71
			✓	85.58	71.43	82.91
num=2	✓	✓		**85.83**	**71.59**	**83.04**
		✓	✓	85.72	71.36	82.9
	✓		✓	85.47	71.18	82.91
num=3	✓	✓	✓	85.5	70.81	82.53

TABLE VI
COMPARISON OF MODEL PARAMETERS, GDLOPS, AND ACCURACY ON THE VAIHINGEN DATASET

Method	Index		Accuracy (%)	
	Params (MB)	GFLOPs	MIou	Average F1
FCN	22.7	5.37	63.46	76.65
UNet	25.13	10.72	66.35	79.13
Deeplab V3+	38.48	20.78	67.16	79.71
UperNet	102.13	26.03	65.84	78.69
DANet	45.36	125.78	65.14	78.00
TransUNet	100.44	35.84	67.11	79.86
Swin-Unet	25.89	7.7	58.01	72.02
UNetFormer	11.72	5.57	69.34	81.51
CMT	57.22	11.85	70.31	82.12
A^2-FPN	12.16	3.16	70.63	82.32
ST-UNet	160.97	78.69	70.23	82.15
SCG-TransNet	41.98	52.97	70.29	82.27
ESMS-Net	328.57	84.55	**71.66**	**83.08**

5) Analysis of the Weight Parameters in MFWF: We analyzed the weight parameters in MFWF on the Vaihingen dataset to obtain the optimal configuration. The parameters λ and η were adjusted to obtain the optimal solution. From Table IV, when λ is set to 0.2 and η to 0.1, the model achieves the best semantic feature fusion performance, with an OA of 85.78%, an MIoU of 71.66%, and an average $F1$ of 83.08%. From the segmentation maps in Fig. 10, when the parameters are set to (0.2, 0.1), the model exhibits the best segmentation performance. Compared with the other parameters, it achieves higher pixel recognition accuracy, greatly reducing the misclassification of the yellow "Car" class.

6) Analysis of Mask: In the auxiliary enhanced encoder, we need to determine which categories of land-cover exhibit the best masking effect. In the Vaihingen dataset, some categories, such as "Impervious Surface," "Building," and "Low Vegetation," have larger coverage and are easily learned during the training process to extract feature information. During iterative learning, we remove the easy segmentation classes based on the segmentation results to further facilitate the learning of hard classes. Table V illustrates the removal of the easy classes to obtain an optimal Mask in ESMS-Net.

ESMS-Net considers the easy segmentation classes including "Impervious Surface," "Building," and "Low Vegetation" as the primary land-covers to be excluded in the Mask. As shown in Table V, we analyze the impact of different Masks for the auxiliary model. From the results, when the Mask selects "Impervious Surface" and "Building," ESMS-Net achieves the highest segmentation accuracy.

7) Efficiency Analysis: To comprehensively compare the efficiency of different networks, as shown in Table VI, we have listed the number of parameters (Params), GFLOPs, average $F1$, and MIoU for all the networks under the same running environment on the Vaihingen dataset. The network parameters and GFLOPs of ESME-Net have larger values than the compared methods, but ESMS-Net has achieved a significant improvement in MIoU and average $F1$. Because the ESMS-Net contains more modules and more complex structures to extract more refined features, resulting in more network parameters and lower computational efficiency. However, compared with the increased accuracies, the computational complexity is acceptable.

V. CONCLUSION

This article introduces a novel semantic segmentation framework, ESMS-Net, which achieves more precise segmentation of land-covers. This network comprises a local–global joint encoder, an auxiliary enhanced encoder, a DFFM, and a MFWF. In the local–global joint encoder, we design a PPAFormer via a pyramid-structured self-attention mechanism to achieve long-range context information interaction. In the decoder, we propose the DFFM to cross-fuse deep semantic information from different scales, enhancing feature representations. Additionally, we introduces MFWF in the decoding stage, which fuses the multilevel decoding features, further improving feature fusion and segmentation performance. In the auxiliary encoder, we construct a semantic mask based on the predicted segmentation results to refine features of hard classes. Through ablation and comparative experiments on the ISPRS Vaihingen and ISPRS Potsdam datasets, the effectiveness of ESMS-Net has been validated. In our network, it needs a large number of labeled samples to train the model. To reduce the cost of labeling samples, we will develop this network as a semi-supervised structure.

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